Feature Importance Analysis

After initial cleaning, 67 predictor features remained. To reduce dimensionality and avoid overfitting, we further selected the top 40 features based on SHAP importance from an initial XGBoost model trained on the full 67-feature set. SHAP was chosen because it provides consistent, model-agnostic importance estimates (Lundberg & Lee, 2017). The 40 features with the highest mean absolute SHAP values were retained for all subsequent modeling. This step reduced noise, improved model interpretability, and maintained high predictive performance (see Model Implementation sections).

Feature importance was evaluated using both XGBoost feature importance and SHAP. SHAP is the primary interpretation tool because it quantifies the direction and magnitude of each feature's effect on predictions, not just its frequency of use. The ranked features from each method are presented below.

1. XGBoost Feature Importance

Top 40 features ranked by XGBoost importance.

Rank	Variable	Description
1	**ADDEPEV3**	(ever told) you had a depressive disorder
2	**DECIDE**	Difficulty concentrating or remembering
3	**EMTSUPRT**	How often get emotional support needed
4	**PHYSHLTH**	Number of days physical health not good
5	**_RFHLTH**	Adults with good or better health
6	**SDHBILLS**	Were you not able to pay your bills?
7	**DIFFALON**	Difficulty doing errands alone
8	**_AGE80**	Imputed age value collapsed above 80
9	**_TOTINDA**	Leisure time physical activity calculated variable
10	**GENHLTH**	General health
11	**SDHUTILS**	Were you not able to pay utility bills or threatened to lose service?
12	**MEDCOST1**	Could not afford to see doctor
13	**_SEX**	Calculated sex variable
14	**SDHFOOD1**	How often did the food that you bought not last, and you didn't have money to get more?
15	**_HISPANC**	Hispanic, Latino/a, or Spanish origin calculated variable
16	**DRNKANY6**	Drink any alcoholic beverages in past 30 days
17	**PNEUVAC4**	Pneumonia shot ever
18	**CHILDREN**	Number of children in household
19	**MARITAL**	Marital status
20	**ALCDAY4**	Days in past 30 had alcoholic beverage
21	**_IMPRACE**	Imputed race/ethnicity value
22	**CVDSTRK3**	Ever diagnosed with a stroke
23	**SDHTRNSP**	Has a lack of reliable transportation kept you from appointments, meetings, work, or getting things needed
24	**EDUCA**	Education level
25	**EXERANY2**	Exercise in past 30 days
26	**_ASTHMS1**	Computed asthma status
27	**_AIDTST4**	Ever been tested for HIV calculated variable
28	**CVDCRHD4**	Ever diagnosed with angina or coronary heart disease
29	**_INCOMG1**	Computed income categories
30	**_MICHD**	Respondents that have ever reported having coronary heart disease (CHD) or myocardial infarction (MI)
31	**PERSDOC3**	Have personal health care provider?
32	**BMI**	Body mass index
33	**HAVARTH4**	Told have arthritis
34	**DEAF**	Are you deaf or do you have serious difficulty hearing?
35	**_STATE**	State FIPS code
36	**_RFBING6**	Binge drinking calculated variable
37	**DROCDY4_**	Computed drink-occasions-per-day

Rank	Variable	Description
38	**EMPLOY1**	Employment status
39	**_SMOKER3**	Computed smoking status
40	**CHECKUP1**	Length of time since last routine checkup

2. SHAP Feature Importance

Top 40 features ranked by SHAP importance.

Rank	Variable	Description
1	**ADDEPEV3**	(ever told) you had a depressive disorder
2	**PHYSHLTH**	Number of days physical health not good
3	**EMTSUPRT**	How often get emotional support needed
4	**DECIDE**	Difficulty concentrating or remembering
5	**_AGE80**	Imputed age value collapsed above 80
6	**GENHLTH**	General health
7	**_SEX**	Calculated sex variable
8	**BMI**	Body mass index
9	**_STATE**	State FIPS code
10	**SDHFOOD1**	How often did the food that you bought not last, and you didn't have money to get more?
11	**DIFFALON**	Difficulty doing errands alone
12	**MARITAL**	Marital status
13	**_INCOMG1**	Computed income categories
14	**SDHBILLS**	Were you not able to pay your bills?
15	**MEDCOST1**	Could not afford to see doctor
16	**CHECKUP1**	Length of time since last routine checkup
17	**_IMPRACE**	Imputed race/ethnicity value
18	**EDUCA**	Education level
19	**SDHUTILS**	Were you not able to pay utility bills or threatened to lose service?
20	**ALCDAY4**	Days in past 30 had alcoholic beverage
21	**INCOME3**	Income level
22	**SDHEMPLY**	Have you lost employment or had hours reduced?
23	**EXERANY2**	Exercise in past 30 days
24	**HHADULT**	Number of adults in household
25	**DROCDY4_**	Computed drink-occasions-per-day
26	**EMPLOY1**	Employment status
27	**DIFFWALK**	Difficulty walking or climbing stairs
28	**_AIDTST4**	Ever been tested for HIV calculated variable
29	**PNEUVAC4**	Pneumonia shot ever
30	**_ASTHMS1**	Computed asthma status
31	**_SMOKER3**	Computed smoking status
32	**_METSTAT**	Metropolitan status
33	**PERSDOC3**	Have personal health care provider?
34	**_RFHLTH**	Adults with good or better health
35	**_RFBING6**	Binge drinking calculated variable
36	**CHILDREN**	Number of children in household
37	**DIABETE4**	(ever told) you had diabetes
38	**FLUSHOT7**	Adult flu shot/spray past 12 months
39	**_HISPANC**	Hispanic, Latino/a, or Spanish origin calculated variable
40	**SDHTRNSP**	Has a lack of reliable transportation kept you from appointments, meetings, work, or getting things needed

3. Interpretation and Feature Selection

Positive SHAP values indicate that a feature pushes the predicted number of mentally unhealthy days higher, while negative SHAP values push it lower. SHAP was used as the primary interpretation method because it measures how strongly each feature affects the model prediction.

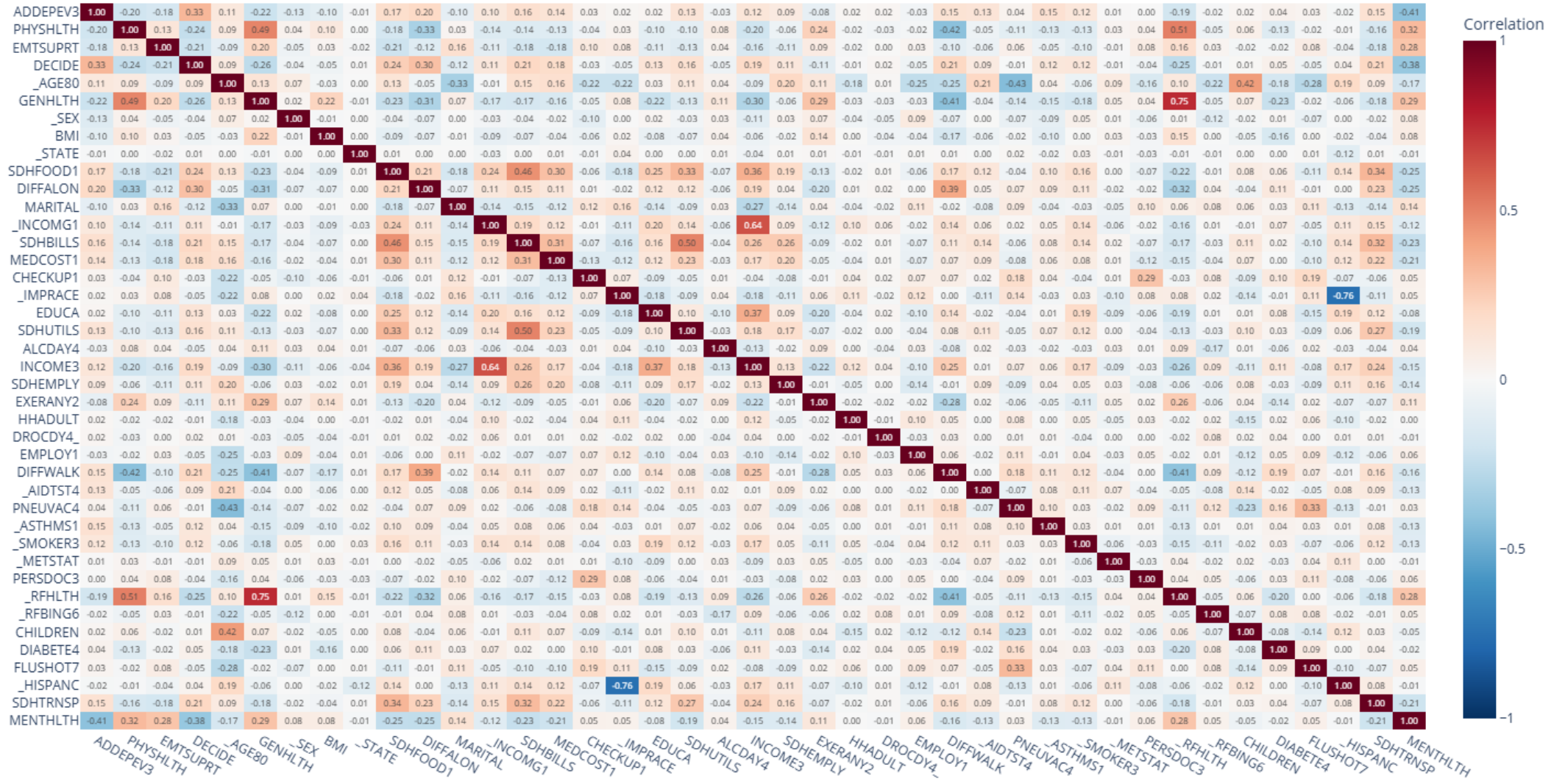
Finally, the most important features identified by SHAP were used to implement the machine learning models.

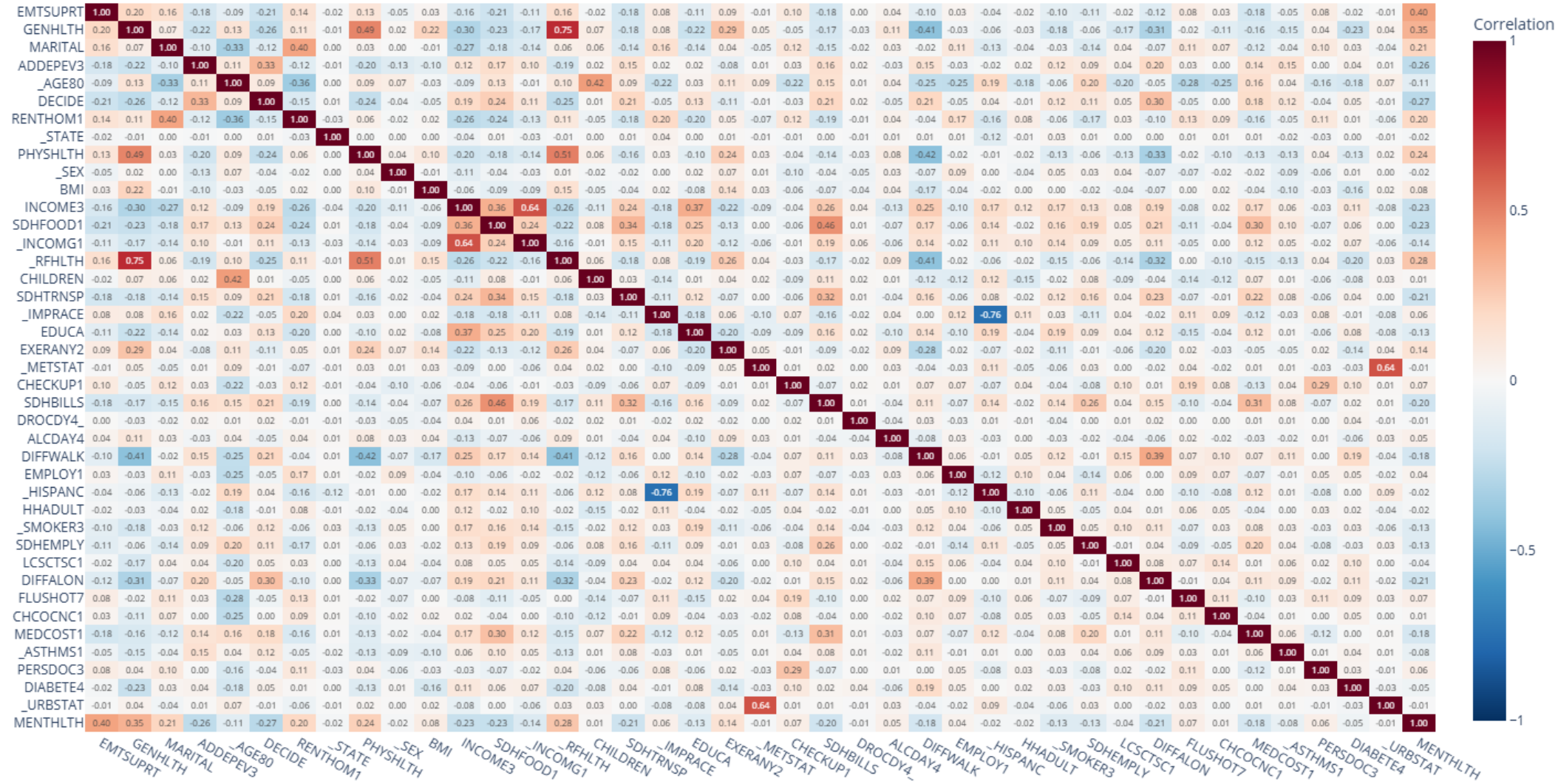
The top 40 features were selected because they collectively accounted for >95% of the total SHAP importance in the initial full-feature model. The remaining 27 features each contributed less than 0.5% to the overall importance and were omitted to simplify the model without sacrificing performance. This threshold is consistent with common practice in applied machine learning and was validated by comparing cross-validation scores between the full (67-feature) and reduced (40-feature) models; the difference in MAE was < 0.05 days, confirming that the reduction is safe.

4. Correlation Heatmap

The feature importances are shown in the heatmaps below.

Correlation Heatmap: Top Features (MENTHLTH) vs Target




Correlation Heatmap: Top Features (LSATISFY) vs Target
EMTSUPRT GENHLTH MARITAL ADDEPEV3 _AGE80 DECIDE RENTHOM1 _STATE PHYSHLTH _SEX BMI INCOME3 SDHFOOD1 _INCOMG1 _RFHLTH CHILDREN SDHTRNSP _IMPRACE EDUCA EXERANY2 _METSTAT CHECKUP1 SDHBILLS DROCDY4_ ALCDAY4 DIFFWALK EMPLOY1 _HISPANC HHADULT _SMOKER3 SDHEMPLY LCSCTSC1 DIFFALON FLUSHOT7 CHCOCNC1 MEDCOST1 _ASTHMS1 PERSDOC3 DIABETE4 _URBSTAT MENTHLTH
Correlation
1
0.5
0
−0.5
−1